№13

$\xi = N(a_\xi\ \sigma_\xi^2)$ $\qquad n = 139$

$\eta = N(a_\eta\ \quad)$ $\qquad m = 108$

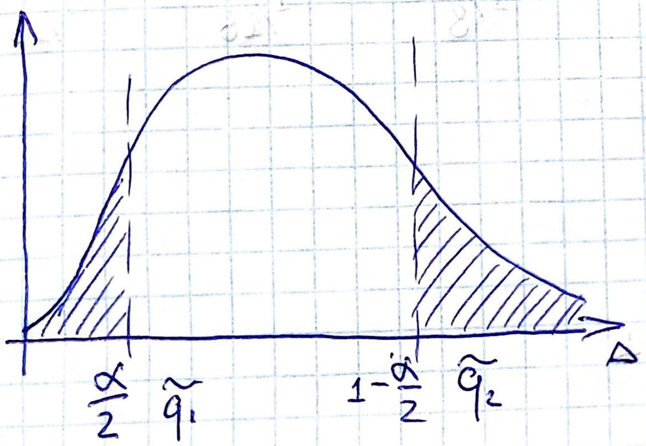
Длина $\quad S_\xi^2 = 5,722 \qquad S_\eta^2 = 6,161$

Ширина $\quad S_\xi^2 = 4,612 \qquad S_\eta^2 = 5,055$

$H_0: \sigma_\xi = \sigma_\eta$ т.е $\dfrac{S_\xi^2}{S_\eta^2} \sim F(n-1\ \ m-1)$

$H_1: \dfrac{\sigma_\xi}{\sigma_\eta} = k \neq 1$ т.ч. $\dfrac{S_\xi^2}{k^2 S_\eta^2} \sim F(n-1\ m-1)$

Тогда



$$\Delta = \left(\frac{S_\xi}{S_\eta}\right)^2$$

$$G: \left[\begin{array}{l} \Delta \geq \tilde{q}_2 \\ \Delta \leq \tilde{q}_1 \end{array}\right.$$

Имеем

$$\alpha_1 = P\left(\left[\begin{array}{l} \Delta \geq \tilde{q}_2 \\ \Delta \leq \tilde{q}_1 \end{array}\right. \middle| H_0\right) = \alpha$$

т.е $\quad \Delta \sim a^2\ F(n-1, m-1)$

– ф-ия распред Фишера

$$W = P\left(\left[\begin{array}{l} \Delta \geq \tilde{q}_2 \\ \Delta \leq \tilde{q}_1 \end{array}\right. \middle| H_1\right) = \hat{f}\left(\frac{\tilde{q}_1}{a^2}\right) - 1 - \hat{f}\left(\frac{\tilde{q}_2}{a^2}\right)$$

где $\hat{f}(x)$ – плотн. распред Фишера